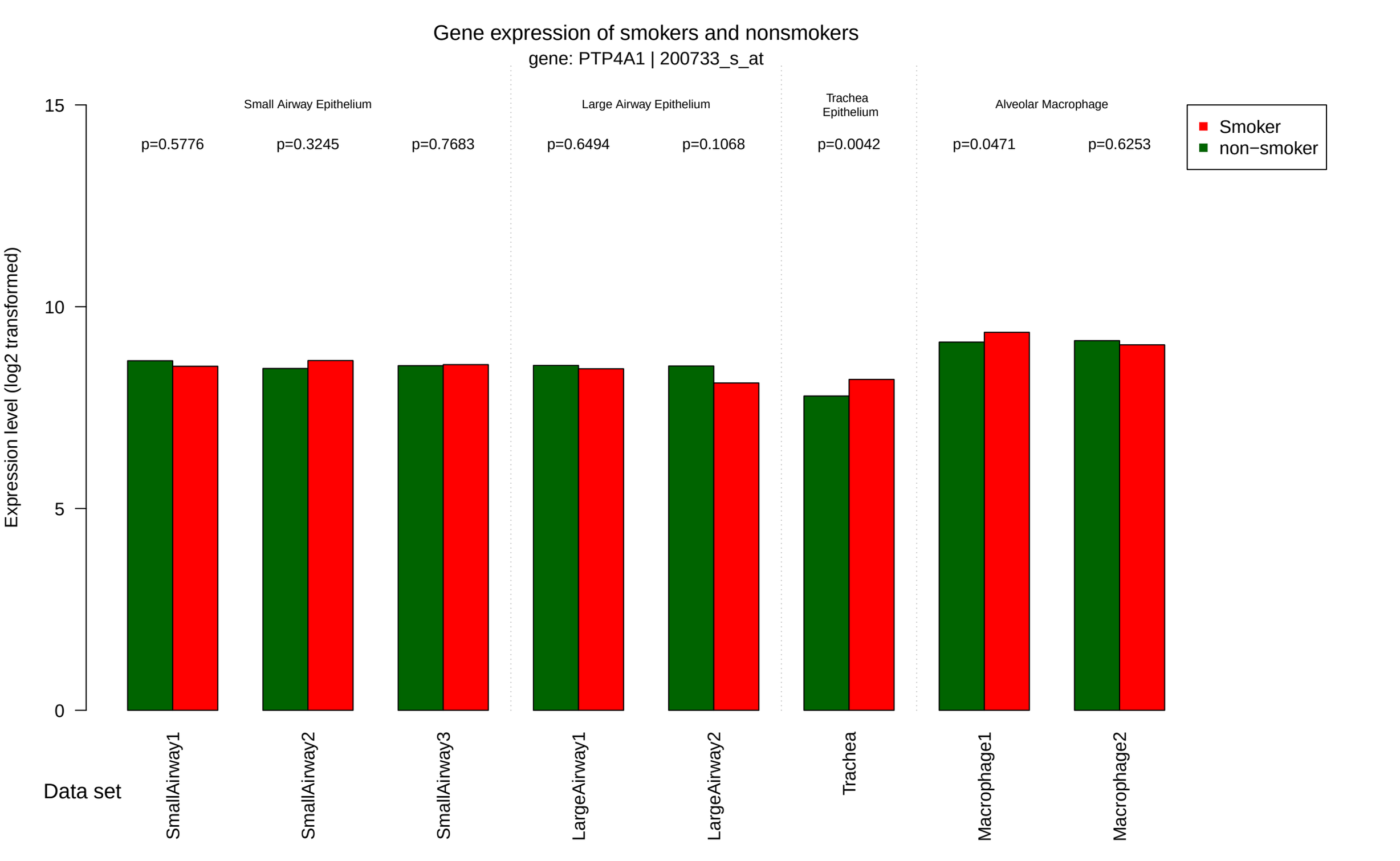
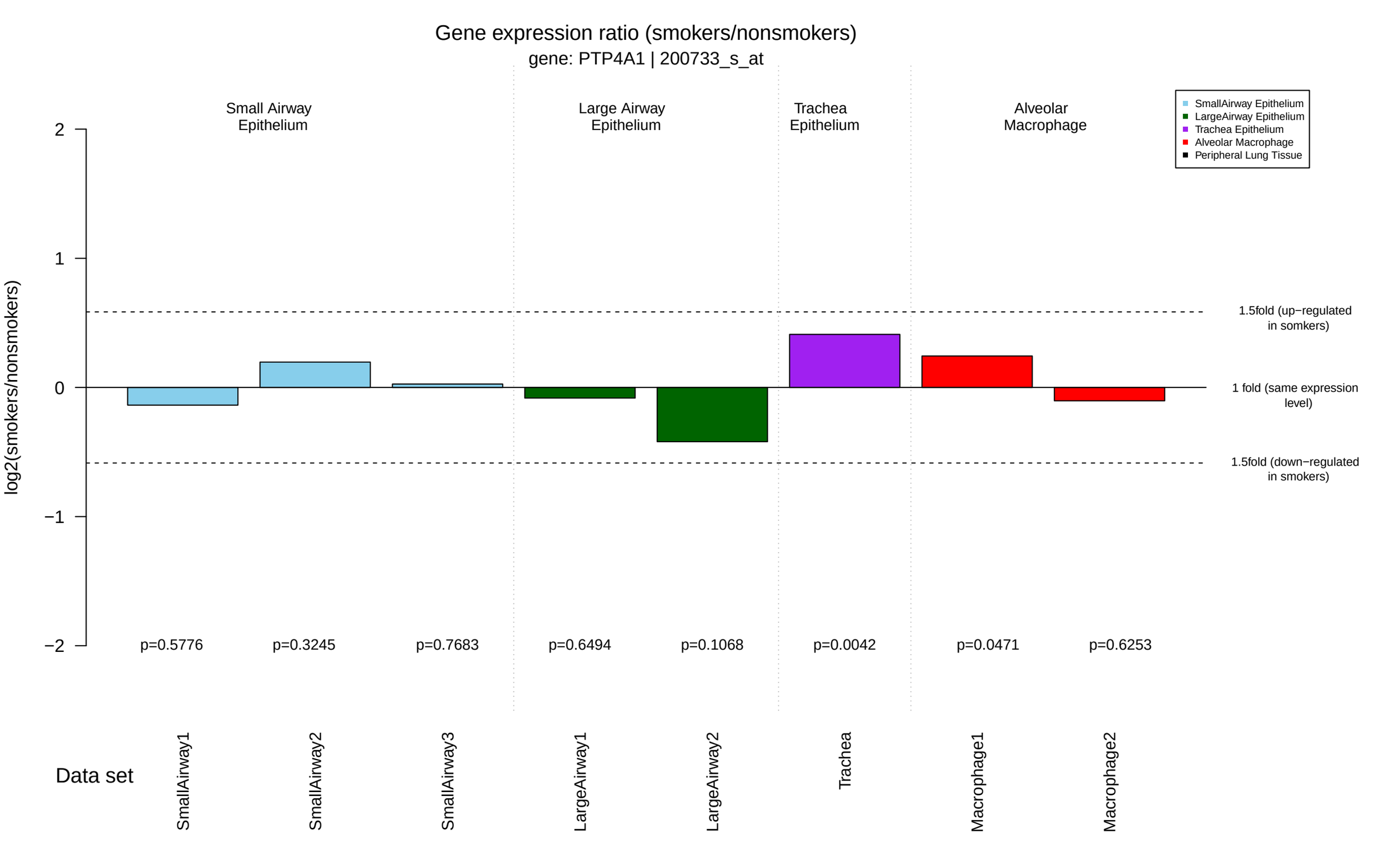
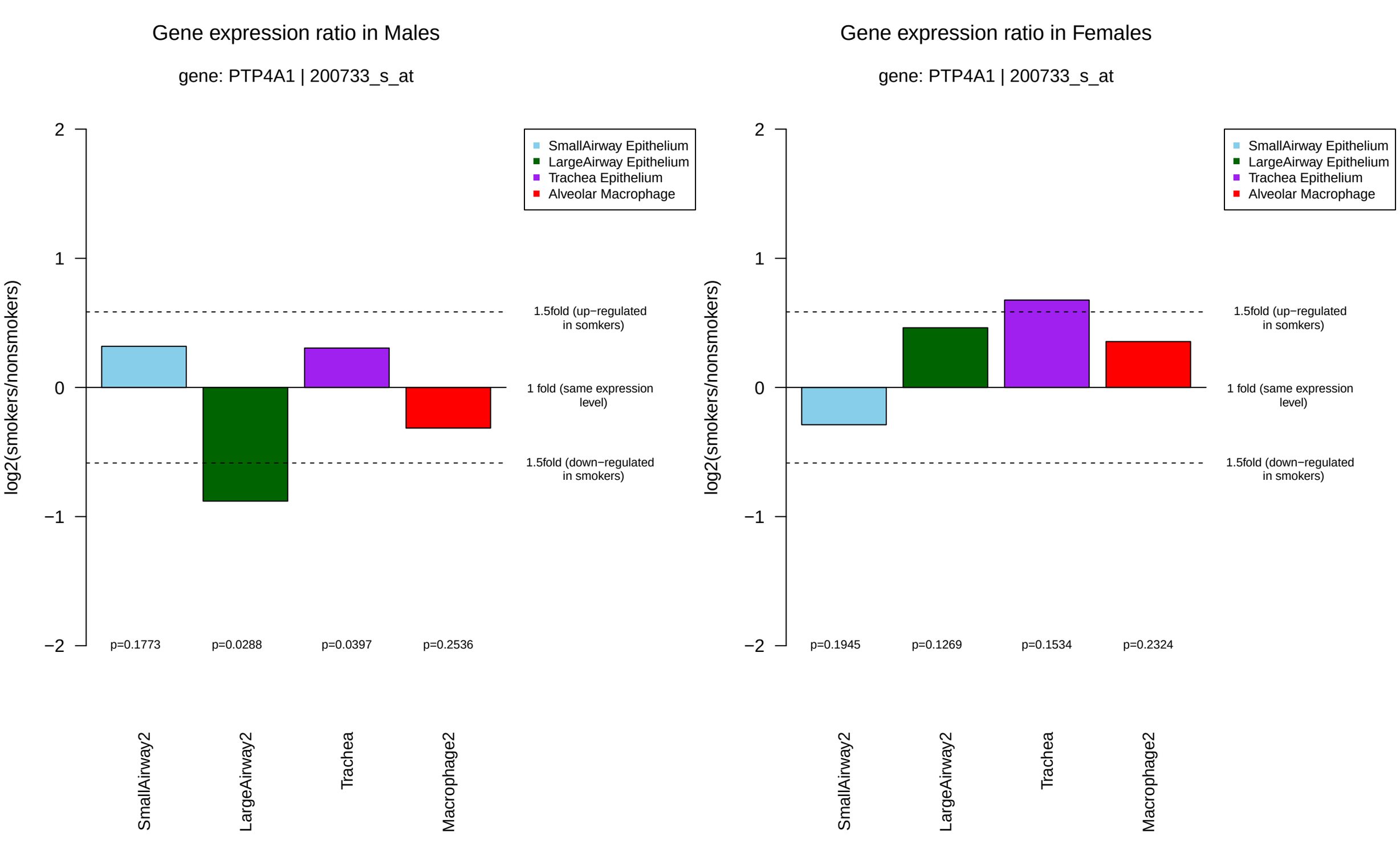
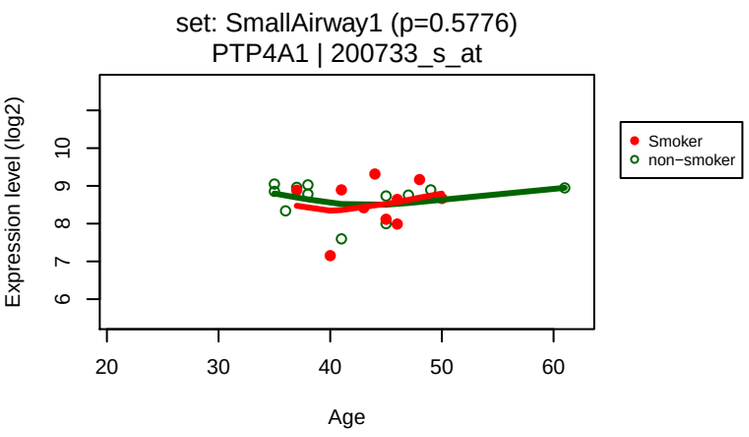
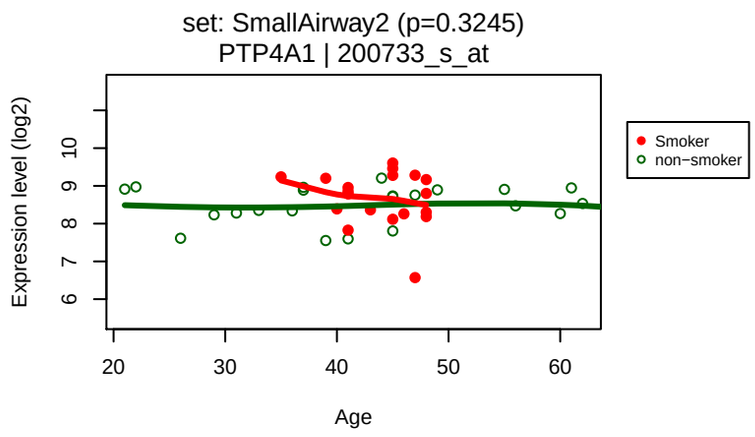
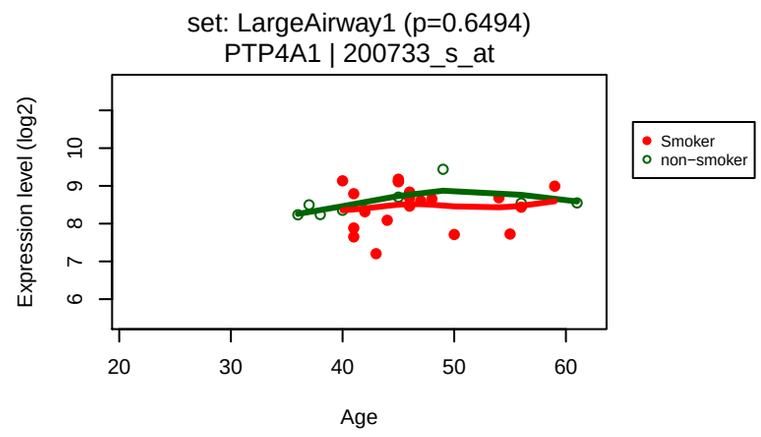

Gene expression of smokers and nonsmokers
gene: PTP4A1 | 200733_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5776
p=0.3245
p=0.7683
p=0.6494
p=0.1068
p=0.0042
p=0.0471
p=0.6253
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

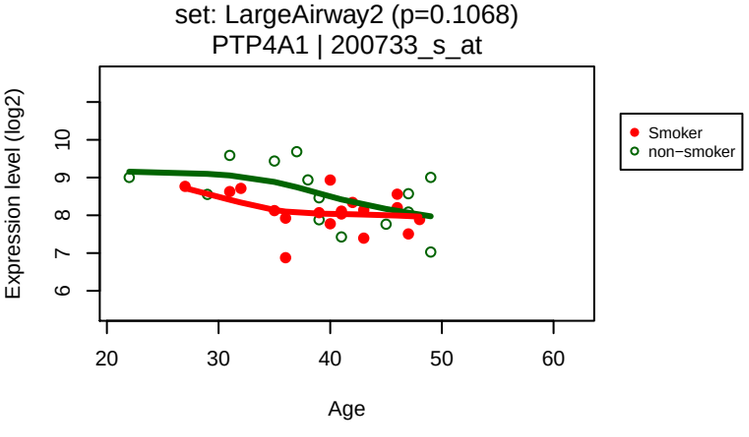

Gene expression ratio (smokers/nonsmokers)
gene: PTP4A1 | 200733_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5776
p=0.3245
p=0.7683
p=0.6494
p=0.1068
p=0.0042
p=0.0471
p=0.6253
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

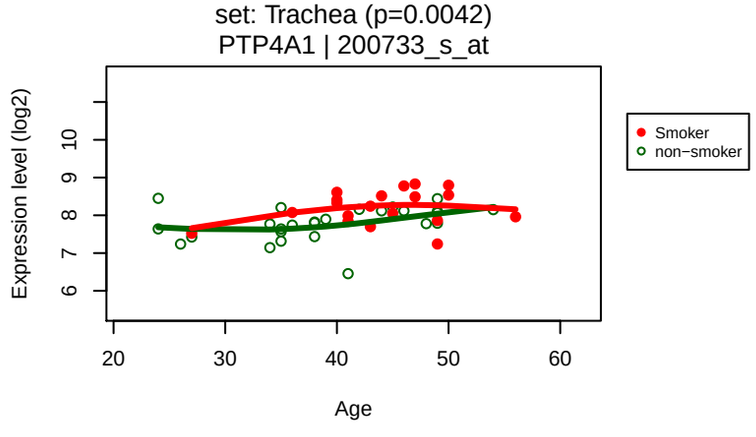

Gene expression ratio in Males
gene: PTP4A1 | 200733_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1773
p=0.0288
p=0.0397
p=0.2536
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTP4A1 | 200733_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1945
p=0.1269
p=0.1534
p=0.2324
SmallAirway2
LargeAirway2
Trachea
Macrophage2

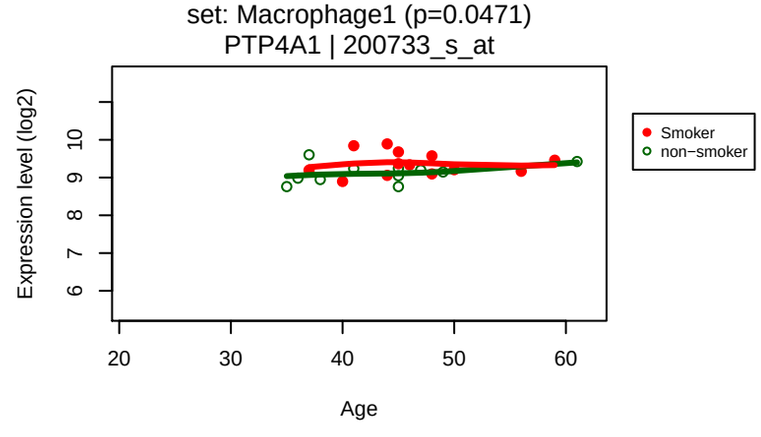

set: SmallAirway1 (p=0.5776)
PTP4A1 | 200733_s_at
Expression level (log2)
Age
Smoker
non−smoker

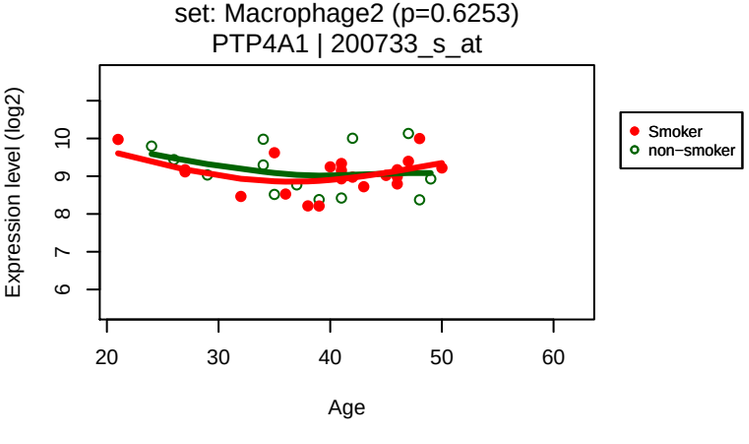

set: SmallAirway2 (p=0.3245)
PTP4A1 | 200733_s_at
Expression level (log2)
Age
Smoker
non−smoker

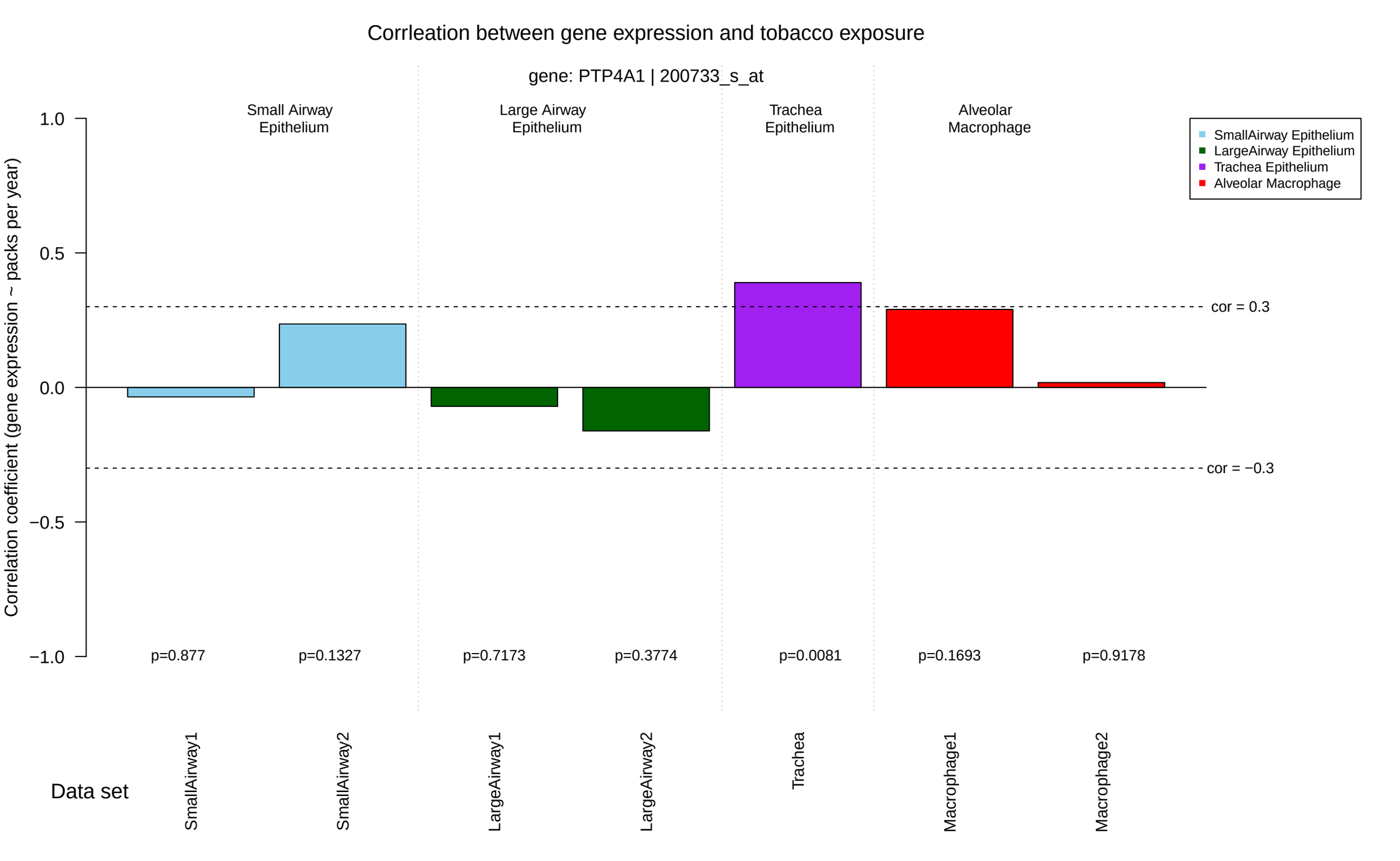

set: LargeAirway1 (p=0.6494)
PTP4A1 | 200733_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1068)
PTP4A1 | 200733_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0042)
PTP4A1 | 200733_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0471)
PTP4A1 | 200733_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6253)
PTP4A1 | 200733_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTP4A1 | 200733_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.877
p=0.1327
p=0.7173
p=0.3774
p=0.0081
p=0.1693
p=0.9178
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2